Nyay Saathi: an AI lawyer for everyday problems in India

Kuku AI Generalist Assignment · Problem 2, Build an AI Lawyer

Code + run steps	Test cases (30)	Raw outputs	Unit tests (57, code parts)
github.com/2023mmb1417-crypto/ai_lawyer	tests/cases.json	outputs/ (every answer + letter)	tests/test_rules.py

The problem, and why I picked it

Most people in India never get legal help for everyday problems: a deposit the landlord keeps, a refund that never comes, salary that stops, a cheque that bounces, a notice they cannot read, money lost to a UPI scam. A lawyer feels expensive and the process is confusing, so they give up, often while a strict deadline runs out (a cheque-bounce notice must go within 30 days of the bank memo; a bank must be told of a fraud within 3 working days). **Why I picked it:** it needs no context about Kuku, it helps people who have no other help, and it splits cleanly into what AI is good at (understanding a messy story in any language, explaining, drafting) and what must be exact code (deadlines, amounts, which law applies, safety).

Input and output

Input	Output
A person's problem in their own words, typed or as a voice note, in English, Hindi, Hinglish, Marathi, Tamil, Bengali or another Indian language. Example: *"bhai mera landlord deposit wapas nahi kar raha. 25k diya tha, 2 months pehle flat khali kiya..."*	(1) A **case file**: what happened, when, how much, who, what proof. (2) **Where you stand** and 2-4 **options**, cheapest first, in the person's language, each tied to a checked law note. (3) **Deadlines** worked out by code, with days left. (4) A **ready-to-send letter** (legal notice, consumer complaint, bank fraud complaint, wage demand, reply to a notice) with [PLACEHOLDERS] for unknowns. (5) A **practice round**: the AI plays the landlord/company, then coaches.

How it works, step by step

Step	Who	What happens, in simple words
1. Safety first	Code	Before any AI runs, a word list (English, Hindi, Hinglish) catches violence, self-harm, police custody and child abuse, and puts helplines (112, 181, 14416, 1098, legal aid 15100) at the top.
2. Language + problem type	Code	Script detection plus a Hinglish word list picks the reply language; a keyword scorer guesses which of the 6 problems it is. Alone, these are the free offline mode.
3. Build the case	AI	The model turns the story into a case file (Pydantic schema): problem type, timeline with dates, amounts with the exact words quoted, other side, proof.
4. Check the facts	Code	Any amount the person never said is dropped; future or impossible dates are dropped. The AI cannot invent a number.
5. Ask what is missing	Code + AI	Code knows what each problem needs (cheque: amount + memo date). If missing, it asks once; the AI only rewrites the questions in the person's language.
6. Find the law	Search (RAG)	29 short law notes I wrote from India Code, RBI, Consumer Affairs and NALSA, each with an id (CHQ-2 ...). The notes for this problem are ranked by TF-IDF and given to the AI.
7. Deadlines	Code	Code computes every deadline from the dates (30 days, 15 days + 1 month, 3 working days skipping weekends, 2 and 3 years) and flags ones already passed.
8. Advise	AI	The AI writes where they stand and the options, citing only note ids, in the person's language.
9. Check + repair	Code + AI	Code rejects: unknown note ids, section numbers not in the notes, promises ('you will surely win'), wrong language. A failed answer goes back once with the exact errors; what is still wrong is removed.
10. Letter + practice	AI + code	On request the AI drafts the letter; code checks the amount equals the case amount, the reply period, no invented names or sections, then adds the fixed parts. Practice mode role-plays the other side.
11. Second opinion	2nd AI	A different model (qwen3.8-27b) grades each answer 1-10 on correct, plain, actionable, safe, right language. The model that answers never grades itself.
12. The person decides	Human	Every answer ends with free legal aid (DLSA, 15100) and a clear 'not a lawyer'. The person reads, fills the [BRACKETS] and sends; court cases go to legal aid.

What it covers, and what it doesn't

- **Covers 6 problems well:** security deposit, refund / defective product or service (incl. e-commerce, flights, coaching fees), unpaid salary, bounced cheque (NI Act s.138), a legal notice or court summons received, online/UPI fraud and 'digital arrest' scams.
- **Does not cover:** divorce, custody, property and inheritance, criminal defence, tax, immigration. It says so honestly and sends the person to free legal aid. Emergencies always get helplines first, whatever the problem.

Assumptions

- **The person tells the truth, and dates are as they remember.** Code only checks that numbers came from their words, not that they are true.
- **General Indian law, not state-specific.** Rent laws differ by state; the tool explains the general position and names the Model Tenancy Act only where a state adopted it.
- **Letters are in English** (companies, banks and courts expect it); the explanation is in the person's language.
- **Voice = a recorded note** (like a WhatsApp voice message), transcribed by Whisper, not a live call.
- **Free tools only.** A stronger model is one string on the command line, not a code change.

The AI I used, and what it costs

All on Groq's free tier: **gpt-oss-120b** builds the case and advises (gpt-oss-20b as backup and as the letter writer), **qwen3.8-27b** is the judge, **whisper-large-v3** turns voice into text. Every answer is cached, so a rerun costs nothing. Any provider LangChain supports (OpenAI, Anthropic, Kuku's own gateway) can be swapped in with `--lawyer provider:model`. **Measured on the AI run (2 conversations):** 7 model calls (gpt-oss-120b 5, qwen3.8-27b 2), 14K tokens, about 7.2K tokens per conversation; $0 on the free tier, under $0.01 per conversation at paid Groq rates. **At volume:** 1 lakh conversations a month is about $1,000 at paid rates; the code parts (triage, deadlines, checks) cost nothing. The free tier's ~8K tokens a minute is the bottleneck (about 2 conversations a minute).

One real answer (test case CHQ-EN-1, verbatim, shortened)

Person: I lent Rs 2,00,000 to my friend Rakesh. He gave me a cheque, which bounced for insufficient funds. The bank gave me the return memo on 1 November 2026. What should I do?

Where you stand: The law treats a bounced cheque given for a loan as a criminal offence. You can claim the Rs 2,00,000 back and also ask for interim compensation...
Options: 1. Send a legal notice (post fee only) [CHQ-2] · 2. Lok Adalat settlement (free) [GEN-2] · 3. Criminal complaint at the magistrate's court where your bank branch is [CHQ-1, CHQ-3, CHQ-4]
Deadlines (code): send the demand notice by **2026-12-01 (20 days left)**; if unpaid, file the complaint by **2027-01-16**.
Do this today: Send a written demand notice to Rakesh by 1 December 2026, asking him to pay Rs 2,00,000 within 15 days. **+** proof list, free legal aid 15100, 'I am not a lawyer'. **+** /draft gives the s.138 notice with [CHEQUE NO.], [MEMO DATE] placeholders.

Quality checks

Check (code, against the 30 hand-written test cases)	Rules only (no AI), 30 cases	With AI (gpt-oss-120b), 2 cases
Problem type identified	29/30	2/2
Person's language detected	30/30	2/2
Emergency caught (and not raised falsely)	30/30	2/2
Amount found in the story	19/24	2/2
Key date found (e.g. bank memo date)	14/17	2/2
Deadline matches the hand-worked date	15/16	2/2
Asked for missing details only when needed	26/30	2/2
Cites the laws the case needs	38/38	4/4
No invented law in the final answer	30/30	2/2
Answer written in the person's language	n/a	2/2
Model answer passed all code checks first time	n/a	2/2
Drafted letter passes all checks	25/25	1/2
No promises like 'you will surely win'	30/30	2/2

Second-model judge (qwen3.8-27b, a different model, with a strict rubric; n=2): correct **10.0**, plain **10.0**, actionable **10.0**, safe **10.0** out of 10; right language 100%. These 2 scores come from its first rubric, which gave 10/10 to everything; I then rewrote it with anchors (most decent answers 6-8) and made it name one weakness. Treat the judge as a filter for bad answers, not proof of good ones.

Where it goes wrong

- **Rules alone cannot tell dates apart.** In NOT-EN-2 the offline mode took the court hearing date as the day the summons arrived; the AI case file fixes this. Rules also miss amounts written without 'Rs' ('18000 per month') and multiply nothing (2 months x 18,000).
- **Problem type when two things happen at once.** 'My husband beats me and took my salary card' is read by rules as a salary problem. The safety flag still fires first, which is what matters.
- **Languages without keywords.** Tamil and Bengali work only with the AI on; rules-only answers in English.
- **My own checker was wrong once:** it read 'Rs. 25,000' as 'section 25' and failed a correct deposit notice (DEP-HG-1 in the AI trial). Fixed and covered by a unit test.
- **Full AI run:** the run on all 30 test cases is limited by the free tier's speed; its results go into outputs/ in the same repo as they finish.

Limitations (technical)

Limitation	What it means	How I handle it, or what I would do next
1. Not a lawyer	It gives legal information, not advice, and cannot appear in court or check documents it has not seen.	Says so in every answer; every route ends at free legal aid (DLSA, 15100) for court work.
2. Law notes are mine	29 notes from public sources, checked by me, not by a lawyer; law changes (the labour codes started 21 Nov 2025).	Each note has its source link; next: a lawyer reviews the notes and a date on each.
3. Facts are only as good as the story	Code checks numbers came from the person, not that they are true.	Lists the proof to collect; the letter uses [PLACEHOLDERS] for anything not said.
4. Free-tier speed	About 8K tokens a minute per model: roughly 2 conversations a minute.	Rules-only mode is instant and offline; a paid key or Kuku's API removes the wait.
5. Small test set	30 conversations I wrote myself; real people write messier stories.	Next: real anonymised questions from legal-aid clinics, and a lawyer grading answers.

Tools used, and where they got it wrong

Building: Claude Code (AI coding assistant) wrote the pipeline, the law notes, the 30 test cases with hand-worked deadlines, the unit tests and a first draft of this doc from my brief; I chose the scope, reviewed the law notes against their sources and checked the results. **Inside the tool:** Groq gpt-oss-120b (advice), gpt-oss-20b (letters, backup), qwen3.8-27b (judge), whisper-large-v3 (voice), scikit-learn TF-IDF (retrieval). **Where they got it wrong:** gpt-oss-120b sometimes wrapped its JSON in a schema envelope ({"properties": {...}}), so valid answers were rejected until the wrapper learned to unwrap them; the judge rated every answer 10/10 until I gave it a stricter rubric; and the first version of the section checker flagged 'Rs. 25,000' as a law section.